

RATIONAL DECISIONS

CHAPTER 16

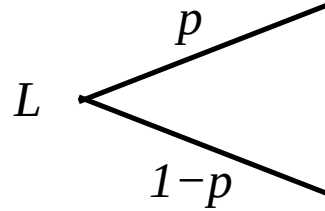
Outline

- ◇ Rational preferences
- ◇ Utilities
- ◇ Money
- ◇ Multiattribute utilities
- ◇ Decision networks
- ◇ Value of information

Preferences

An agent chooses among prizes (A , B , etc.) and lotteries, i.e. prizes with uncertain prizes

Lottery $L = [p, A; (1 - p), B]$



Notation:

$A \succ B$	A preferred to B
$A \sim B$	indifference between A and B
$A \not\succ B$	B not preferred to A

Rational preferences

Idea: preferences of a rational agent must obey constraints.

Rational preferences $\Rightarrow$

behavior describable as maximization of expected utility

Constraints:

Orderability

$$(A \succ B) \vee (B \succ A) \vee (A \sim B)$$

Transitivity

$$(A \succ B) \wedge (B \succ C) \Rightarrow (A \succ C)$$

Continuity

$$A \succ B \succ C \Rightarrow \exists p \ [p, A; 1 - p, C] \sim B$$

Substitutability

$$A \sim B \Rightarrow [p, A; 1 - p, C] \sim [p, B; 1 - p, C]$$

Monotonicity

$$A \succ B \Rightarrow (p \geq q \Leftrightarrow [p, A; 1 - p, B] \succsim [q, A; 1 - q, B])$$

Rational preferences contd.

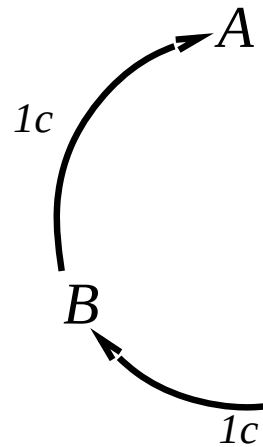
Violating the constraints leads to self-evident irrationality

For example: an agent with intransitive preferences can be induced to give away all its money

If $B \succ C$, then an agent who has C would pay (say) 1 cent to get B

If $A \succ B$, then an agent who has B would pay (say) 1 cent to get A

If $C \succ A$, then an agent who has A would pay (say) 1 cent to get C



Maximizing expected utility

Theorem (Ramsey, 1931; von Neumann and Morgenstern, 1944)

Given preferences satisfying the constraints
there exists a real-valued function U such that

$$U(A) \geq U(B) \iff A \succsim B$$
$$U([p_1, S_1; \dots; p_n, S_n]) = \sum_i p_i U(S_i)$$

MEU principle:

Choose the action that maximizes expected utility

Note: an agent can be entirely rational (consistent with MEU)
without ever representing or manipulating utilities and probabilities

E.g., a lookup table for perfect tictactoe

Utilities

Utilities map states to real numbers. Which numbers?

Standard approach to assessment of human utilities:

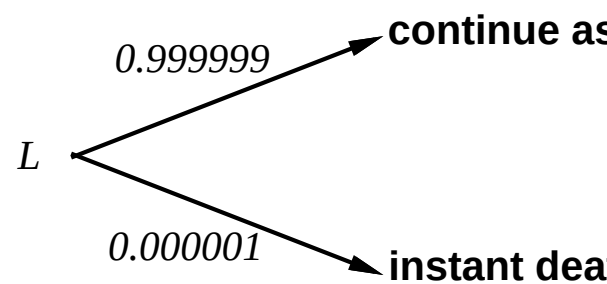
compare a given state A to a standard lottery L_p that has

“best possible prize” $u_{\top}$ with probability p

“worst possible catastrophe” $u_{\perp}$ with probability $(1 - p)$

adjust lottery probability p until $A \sim L_p$

pay \$30 $\sim$



Utility scales

Normalized utilities: $u_{\top} = 1.0$, $u_{\perp} = 0.0$

Micromorts: one-millionth chance of death
useful for Russian roulette, paying to reduce product risks,

QALYs: quality-adjusted life years
useful for medical decisions involving substantial risk

Note: behavior is invariant w.r.t. +ve linear transformation

$$U'(x) = k_1 U(x) + k_2 \quad \text{where } k_1 > 0$$

With deterministic prizes only (no lottery choices), only ordinal utility can be determined, i.e., total order on prizes

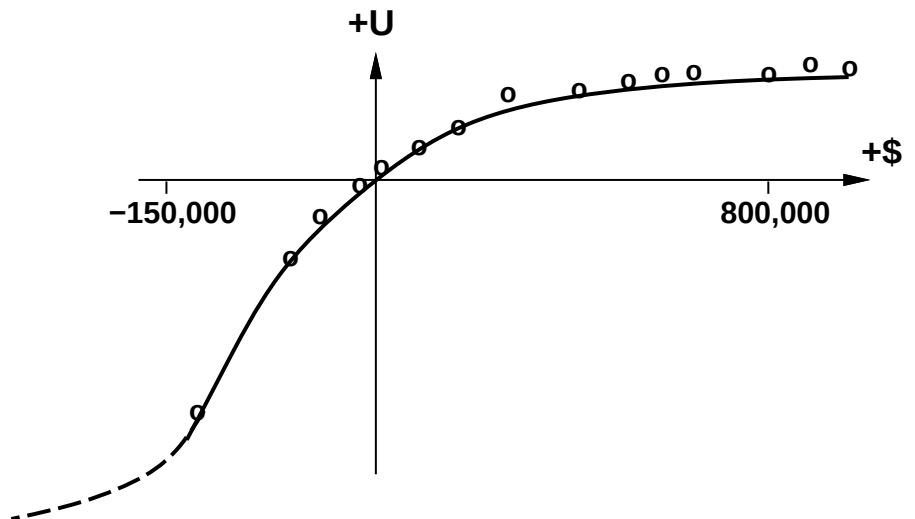
Money

Money does not behave as a utility function

Given a lottery L with expected monetary value $EMV(L)$, usually $U(L) < U(EMV(L))$, i.e., people are risk-averse

Utility curve: for what probability p am I indifferent between a
and a lottery $[p, \$M; (1 - p), \$0]$ for large M ?

Typical empirical data, extrapolated with risk-prone behavior:



Student group utility

For each x , adjust p until half the class votes for lottery ($M=1$)

